

ONLINE RETAIL SALES ANALYSIS

Data Analysis Report

Analysis Period	December 2010 – December 2011
Tools Used	Python, Pandas, Power BI
Analysis Areas	Sales Performance, Product Analysis, Customer Analysis, RFM Segmentation, Geographic Analysis
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1. Executive Summary

This project analyzes historical online retail transaction data to understand overall sales performance, customer behavior, product performance, and geographic revenue distribution.

The analysis began with data cleaning and preparation in Python, followed by exploratory data analysis (EDA), product analysis, customer revenue analysis, RFM-based customer segmentation, geographic analysis, and finally the development of a Power BI dashboard.

The analysis identified several important business patterns:

- Total revenue from the cleaned sales dataset was **£10.28 million**.
- The dataset contained **20,556 orders, 5,645,437 units, 3,931 products, and 38 countries**.
- Identifiable customers generated approximately **£8.77 million** in revenue.
- Customer value was highly concentrated, with the top 10 customers contributing approximately **17.40%** of identifiable customer revenue.
- **High-Value Active customers** generated approximately **64.75%** of identifiable customer revenue.
- The **United Kingdom** was the dominant market.
- A group of **High-Value Lapsed customers** generated approximately **£425K**, making them a potential re-engagement opportunity.
- Product analysis identified several major revenue-generating products, while extreme transactions demonstrated the importance of validating unusual records before drawing conclusions.

RFM analysis was used as one part of the broader customer analysis rather than as the sole focus of the project.

2. Introduction

The retailer has a large volume of transaction-level data containing information about invoices, products, quantities, prices, customers, dates, and countries.

However, raw transaction records do not directly answer important business questions such as:

- How much revenue is being generated?
- How does revenue change over time?
- Which products contribute the most revenue?
- Which customers are the most valuable?
- Which customers are active, recent, frequent, or lapsed?
- Which countries contribute the most revenue?
- Where should the business focus its customer-retention efforts?
- Are there unusual transactions that could significantly affect the analysis?

This project transforms the raw transaction data into structured business insights through data cleaning, exploratory analysis, customer segmentation, and dashboard visualization.

3. Business Problem

The analysis was designed around several connected business problems.

3.1 Sales Performance

Understand the overall scale and performance of the business through:

- Total revenue
- Total orders
- Total quantity sold
- Average Order Value
- Monthly revenue trends

3.2 Product Performance

Identify products that contribute significantly to revenue and investigate whether high revenue is driven by repeated sales, high prices, or unusually large transactions.

3.3 Customer Analysis

Understand customer revenue contribution and identify whether revenue is evenly distributed or concentrated among a smaller group of customers.

3.4 Customer Segmentation and RFM

Use Recency, Frequency, and Monetary analysis to classify customers into meaningful groups such as:

- High-Value Active
- High-Value Lapsed
- Recent Customers
- High-Spend Customers
- Frequent Customers
- Other

The purpose is to understand customer value and behavior and identify groups requiring different business attention. Also which segment brought the most revenue.

3.5 Geographic Performance

Analyze revenue and customer segments across countries to determine which markets contribute the most business value. The most active ones contributed the most.

3.6 Overall Business Opportunity

Combine sales, product, customer, RFM, and geographic findings to identify areas for customer retention, re-engagement, market focus, and further investigation. Which would help us to improve the average order value.

4. Dataset Overview

The original dataset contained:

541,909 rows × 8 columns

The columns were:

Column	Description
InvoiceNo	Invoice / transaction identifier
StockCode	Product identifier
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Transaction date and time
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country

The dataset represents transaction-level retail activity.

The analysis is based on a single transaction table, there is no separate customer, product, inventory, or operational reference table available for independent validation.

5. Data Cleaning and Preparation

5.1 Duplicate Records

A total of **5,268 duplicate rows** were identified and removed. After duplicate removal, the dataset contained **536,641 rows**. (541,909 initially with errors)

Duplicate transaction rows were removed because they represented repeated records rather than independent business activity.

Importantly, repeated dates were not automatically treated as duplicates because the same customer can purchase multiple products within the same order or on the same date.

5.2 UnitPrice Cleaning

The UnitPrice column initially contained text-formatted values, including values containing currency symbols and spaces. The values were cleaned and converted into numeric format. The final UnitPrice data type was **float64**.

Zero-price transactions were retained because there was insufficient evidence to classify them as invalid transactions.

5.3 InvoiceDate Conversion

InvoiceDate was converted into a proper datetime format. The final transaction period was **1 December 2010 – 9 December 2011**.

This conversion allowed the analysis to calculate:

- Monthly revenue
- Order trends
- Customer recency
- Time-based business patterns

5.4 Missing Customer IDs

A significant number of transactions did not contain CustomerID. Instead of removing these transactions, the missing values were replaced with **"Unknown"**.

This preserved the transaction-level sales information while preventing missing values from causing problems during analysis.

Customer-level analysis was performed using identifiable customers, excluding the Unknown customer group where appropriate.

Later they were labelled as "0" for calculations , before this wrong amount of customers were calculated by Power BI.

5.5 Missing Product Descriptions

Missing product descriptions were replaced with **"Unknown_Item"**. This allowed the transactions to remain in the dataset without inventing product descriptions.

5.6 Returns and Sales

Negative-quantity transactions were treated as returns and separated from positive sales transactions. The resulting datasets were:

Dataset	Rows
Sales transactions	529,083
Returns	10,067

The sales dataset was used for the main revenue and sales analysis.

While preserving the original sales data.

5.7 Revenue Calculation

A new Revenue column was created using:

Revenue = Quantity × UnitPrice

The final sales dataset contained **529,083 rows × 9 columns**. The final columns were:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country
- Revenue

6. Treatment of Extreme Transactions

Statistical outlier analysis was performed using the IQR method. The analysis identified unusually high values in:

- Quantity
- UnitPrice
- Revenue

Examples included:

Product	Quantity	Unit Price	Revenue
PAPER CRAFT , LITTLE BIRDIE	80,995	£2.08	£168,469.60
MEDIUM CERAMIC TOP STORAGE JAR	74,215	£1.04	£77,183.60
PICNIC BASKET WICKER 60 PIECES	60	£649.50	£38,970.00

These records are statistically unusual, but statistical unusualness does not automatically mean the transactions are incorrect.

Since only one transaction table was available, there was no separate source that could be used to verify whether these transactions were genuine business activity.

Therefore:

Extreme transactions were retained and were not replaced or removed solely because they were identified as outliers.

This preserves the original business records under the project's assumption that the transactions are valid unless there is independent evidence to prove otherwise.

7. Overall Sales Performance

The cleaned sales dataset produced the following overall KPIs:

KPI	Value
Total Revenue	£10,279,039.89
Total Quantity	5,645,437
Total Orders	20,556
Customers	4,337
Identifiable Customers	4,336
Products	3,931
Countries	38
Average Order Value	£500.05

The difference between total customers and identifiable customers is caused by transactions where CustomerID was unavailable.

8. KPI Calculation Logic

The Power BI KPIs were designed around business definitions rather than simple row counts.

8.1 Total Revenue

```
SUM(Revenue)
```

This represents the total revenue generated by the sales transactions.

8.2 Total Quantity

```
SUM(Quantity)
```

This represents the total number of units sold.

8.3 Total Orders

```
DISTINCTCOUNT(InvoiceNo)
```

This is important because one invoice can contain multiple product rows.

For example, if one customer purchases five different products in one order, the transaction may contain five rows but should count as **1 order** rather than 5 orders. Therefore, distinct InvoiceNo is appropriate for calculating total orders.

8.4 Customers

```
DISTINCTCOUNT(CustomerID)
```

For customer-level analysis, identifiable CustomerIDs are considered. Unknown CustomerIDs can be excluded when calculating identifiable customer metrics.

8.5 Products

```
DISTINCTCOUNT(StockCode)
```

This represents the number of unique product codes.

8.6 Average Order Value

```
Total Revenue ÷ Total Orders
```

This represents the average amount of revenue generated per order.

9. Sales Trend Analysis

Revenue was analyzed by month to understand how business activity changed over time. Some key monthly results were:

Month	Revenue	Orders	Quantity	AOV
Dec 2010	£778,623.36	1,624	361,839	£479.45
Jan 2011	£672,207.41	1,115	396,363	£602.88
Feb 2011	£509,151.42	27,048	286,391	£18.82
Mar 2011	£691,890.92	35,727	383,511	£19.37
Apr 2011	£516,718.21	29,042	311,885	£17.79
May 2011	£741,690.55	36,116	398,280	£20.54
Jun 2011	£739,426.88	35,869	393,967	£20.61

The monthly revenue analysis shows that revenue fluctuates considerably across the period.

The purpose of the monthly trend is not simply to identify the highest month, but to understand whether revenue is stable, seasonal, or affected by particular periods of stronger or weaker activity.

10. Product Performance Analysis

Product-level analysis was used to identify the products contributing the most revenue. Some of the highest-revenue products included:

Product	Revenue
REGENCY CAKESTAND 3 TIER	£174,484.74
PAPER CRAFT , LITTLE BIRDIE	£168,469.60
WHITE HANGING HEART	£106,292.77

The product analysis demonstrates that revenue can be generated through different combinations of:

- Sales volume
- Unit price
- Number of orders
- Large individual transactions

For example, PAPER CRAFT , LITTLE BIRDIE generated £168,469.60 through an exceptionally large transaction.

Therefore, product rankings should be interpreted together with transaction volume and unusual-value investigation.

11. Customer Revenue Analysis

Customer-level analysis was performed using identifiable CustomerIDs. Key results:

Metric	Value
Identifiable Customers	4,336
Identifiable Revenue	£8,767,917.65
Mean Customer Revenue	£2,022.12
Median Customer Revenue	£668.08
Maximum Customer Revenue	£279,138.02
Top 10 Customer Revenue	£1,525,341.56
Top 10 Revenue Share	17.40%

The mean customer revenue is substantially higher than the median. This indicates that customer revenue is highly uneven.

A relatively small number of customers contribute significantly more revenue than the typical customer. This makes customer segmentation and retention analysis particularly important.

12. RFM Analysis

RFM analysis was used to understand customer behavior through three dimensions.

12.1 Recency

Measures how recently a customer purchased.

Lower Recency = more recent customer

12.2 Frequency

Measures how often the customer purchased.

Higher Frequency = more repeat purchasing

12.3 Monetary

Measures how much revenue the customer generated.

Higher Monetary = higher customer value

12.4 RFM Statistics

The RFM analysis date was **9 December 2011 at 12:50**. Key statistics:

Metric	Value
Average Recency	91.69 days
Median Recency	50 days
Average Frequency	4.25
Median Frequency	2
Maximum Frequency	206

RFM provided a structured way to distinguish customers based on both behavior and value.

13. Customer Segmentation

The resulting customer segments were:

Segment	Customers	Revenue
High-Value Active	945	£5,676,823.41
Other	2,256	£1,791,668.65
High-Spend Customers	260	£558,896.33
High-Value Lapsed	173	£425,205.54
Recent Customers	506	£198,423.23
Frequent Customers	196	£116,900.49

13.1 High-Value Active Customers

The High-Value Active segment contained **945 customers** and generated **£5,676,823.41**. This represents approximately **64.75% of identifiable customer revenue**.

This is the most important customer segment from a revenue perspective. These customers demonstrate strong monetary value while remaining relatively active.

13.2 High-Value Lapsed Customers

The High-Value Lapsed segment contained **173 customers** and generated approximately **£425K** historically.

These customers are important because they have already demonstrated significant value but are no longer as recently active. This creates a potential customer re-engagement opportunity.

13.3 Recent Customers

The Recent Customers segment contained **506 customers** and generated **£198,423.23**. These customers may represent an opportunity for developing repeat purchasing behavior.

13.4 High-Spend Customers

The High-Spend Customers segment contained **260 customers** and generated **£558,896.33**. These customers demonstrate high monetary value even if they do not necessarily have the same activity profile as High-Value Active customers.

13.5 Frequent Customers

The Frequent Customers segment contained **196 customers** and generated **£116,900.49**. Their primary characteristic is repeated purchasing behavior.

13.6 Other Customers

The Other segment was the largest by customer count, with **2,256 customers**. However, its revenue contribution was lower relative to the High-Value Active segment.

This demonstrates why customer distribution should not be evaluated using customer count alone.

14. Geographic Analysis

The dataset contains **38 countries**. The United Kingdom was the dominant market.

Examples of customer-segment results include:

Country / Segment	Customers	Revenue
UK — High-Value Active	853	£4,569,998.10
UK — Other	2,053	£1,583,381.60
UK — High-Spend	199	£422,798.89
Netherlands — High-Value Active	2	£280,419.06
EIRE — High-Value Active	2	£257,830.35

These results show that customer count and revenue should be evaluated together.

A country with fewer customers can still generate substantial revenue if those customers have high monetary value.

15. Customer-Country Assignment

During customer-level geographic analysis, eight customers were found to be associated with multiple countries:

12370, 12394, 12417, 12422, 12429, 12431, 12455, 12457

For consistent customer-level analysis, one country was assigned to each customer. This resulted in **4,336 identifiable customers with an assigned country**.

Examples include:

CustomerID	Assigned Country
12346	United Kingdom
12347	Iceland
12348	Finland
12349	Italy
12350	Norway

16. Power BI Dashboard

The Power BI dashboard was designed to convert the analysis into an interactive business view. The dashboard focuses on two major questions.

16.1 Where is the revenue coming from?

This is addressed through:

- KPI cards
- Monthly revenue
- Revenue by country
- Product revenue

16.2 Who are the valuable customers?

This is addressed through:

- Customer segment distribution
- Segment revenue
- Customer revenue analysis
- Country × customer segment analysis
- RFM-based customer segmentation

17. Power BI Visual Explanations

17.1 KPI Cards

Visual	KPI Cards
Displays	Total Revenue, Total Orders, Customers, Products, Total Quantity, Average Order Value

What it shows. The KPI cards provide a high-level snapshot of the business.

Why it is used. A viewer can immediately understand the scale of the business without reading through multiple charts.

Business interpretation. The KPI section provides the baseline for interpreting the rest of the dashboard. For example, the AOV of £500.05 gives context to the number of orders and total revenue.

17.2 Monthly Revenue Trend

Visual	Line Chart
X-axis	Month
Y-axis	Revenue

What it shows. The chart shows how revenue changes over time.

Why a line chart? Time is sequential, so a line chart makes trends, peaks, declines, and fluctuations easy to see.

How to interpret it.

- Upward movement → increasing revenue
- Downward movement → decreasing revenue
- Peaks → stronger revenue periods
- Dips → weaker revenue periods

Business insight. The chart helps identify periods requiring further investigation. For example, strong months may be associated with seasonal demand or increased customer activity, while weaker months may require investigation.

The chart identifies patterns but does not by itself prove why those patterns occurred.

17.3 Revenue by Country

Visual	Horizontal Bar Chart
Y-axis	Country
X-axis	Revenue
Sort	Descending

What it shows. The chart ranks countries according to total revenue.

Why a horizontal bar chart? Country names can be relatively long, making horizontal bars easier to read.

How to interpret it. Longer bars represent greater revenue contribution.

Business insight. The United Kingdom is the dominant market. The chart also helps identify smaller markets that may have disproportionately high revenue.

17.4 Customer Segment Distribution

Visual	Bar / Column Chart
Axis	Customer Segment
Value	Customer Count

What it shows. The chart shows how many customers belong to each segment.

Why it is used. Customer count answers: "**How large is each customer group?**"

Key result. The largest segment is **Other — 2,256 customers**, followed by **High-Value Active — 945 customers**.

Business insight. A large customer segment does not automatically mean a large revenue contribution. Therefore, this chart must be compared with the Segment Revenue chart.

17.5 Revenue by Customer Segment

Visual	Bar / Column Chart
Axis	Customer Segment
Value	Total Revenue

What it shows. The chart measures the financial contribution of each customer segment.

Why it is used. This answers: "**Which customer group generates the most business value?**"

Key result. High-Value Active customers generated **£5,676,823.41**, or approximately **64.75% of identifiable customer revenue**.

Business insight. This is one of the strongest findings in the analysis. The segment is not the largest customer group, but it is by far the largest contributor to revenue. This demonstrates the difference between:

Customer volume ≠ Customer value

17.6 Country × Customer Segment

Visual	Stacked Bar Chart / Grouped Bar Chart / Matrix
Dimensions	Country + Customer Segment
Measure	Revenue or Customer Count

What it shows. This visual combines geography with customer behavior.

Why it is used. It allows the business to understand:

- Which countries contain the most valuable customers
- Which customer segments dominate each country
- Whether revenue is concentrated geographically within particular segments

Business insight. The UK has a particularly strong High-Value Active customer base. At the same time, smaller markets such as the Netherlands and EIRE demonstrate that low customer count does not necessarily mean low business value.

17.7 Top Products by Revenue

Visual	Horizontal Bar Chart
Dimension	Product
Measure	Revenue

What it shows. The visual ranks products by total revenue.

Why it is used. It provides an immediate view of the products contributing the most revenue.

Key results. Top revenue-generating products include:

1. REGENCY CAKESTAND 3 TIER — £174,484.74
2. PAPER CRAFT , LITTLE BIRDIE — £168,469.60
3. WHITE HANGING HEART — £106,292.77

Business insight. The chart helps identify important products, but product revenue should be interpreted alongside order volume and unusual transactions. For example, the PAPER CRAFT product's high revenue is heavily influenced by an exceptionally large transaction.

17.8 Customer Revenue Distribution

Visual	Customer Revenue Distribution / Bar or Distribution Visual
--------	--

What it shows. The visual demonstrates how revenue is distributed across customers.

Why it is used. Average revenue alone can be misleading. For example, mean customer revenue is **£2,022.12**, while median customer revenue is **£668.08**. The large difference indicates that some customers contribute substantially more revenue than the typical customer.

Business insight. Customer value is highly uneven. This supports the use of customer segmentation and RFM analysis to identify high-value customers rather than treating every customer equally.

18. Key Business Insights

18.1 Revenue is concentrated among high-value customers

High-Value Active customers generated approximately **64.75% of identifiable customer revenue**. This makes them the most important customer group from a revenue perspective.

18.2 Customer value is highly uneven

Mean customer revenue was **£2,022.12**, while median customer revenue was **£668.08**. This indicates that a relatively small group of customers contributes disproportionately more revenue.

18.3 The top 10 customers are significant

The top 10 customers generated **£1,525,341.56**, which represents **17.40%** of identifiable customer revenue. This demonstrates meaningful revenue concentration among a small number of customers.

18.4 High-Value Lapsed customers represent a potential opportunity

The High-Value Lapsed group generated **£425,205.54** historically. Because these customers have demonstrated high value but lower recent activity, they may be suitable for re-engagement strategies.

18.5 The UK is the dominant market

The UK contains a large share of the identifiable customer base and revenue. The UK High-Value Active segment alone generated approximately **£4.57 million**.

18.6 Smaller markets can still be valuable

The Netherlands and EIRE demonstrate that customer count alone does not determine market value. For example, **Netherlands High-Value Active: 2 customers → £280,419.06**.

This shows why geographic analysis should combine customer count with revenue.

18.7 Product rankings require transaction context

Some products have high revenue because of exceptionally large individual transactions. Therefore, revenue rankings should be evaluated together with:

- Quantity
- Number of orders
- Unit price
- Transaction-level outliers

19. Business Recommendations

Based on the analysis, the following actions can be considered.

19.1 Protect High-Value Active Customers

Because this group generates the largest share of identifiable customer revenue, customer retention should be a priority. Possible actions include:

- Loyalty programs
- Personalized offers
- Priority customer service
- Repeat-purchase incentives

19.2 Re-engage High-Value Lapsed Customers

The High-Value Lapsed group represents customers who previously demonstrated significant value. Possible actions include:

- Personalized reactivation campaigns
- Product recommendations
- Targeted discounts
- Reminder campaigns

19.3 Develop Recent Customers

Recent customers can be encouraged to make additional purchases. Possible actions include:

- Cross-selling
- Product recommendations
- Follow-up communication
- Second-purchase incentives

19.4 Monitor Revenue Concentration

Since a relatively small number of customers contribute substantial revenue, the business should monitor customer concentration. High dependence on a small customer group may create risk if those customers stop purchasing.

19.5 Investigate High-Value Smaller Markets

Countries with small customer counts but high revenue should be investigated further. The analysis suggests that some markets may contain particularly valuable customers despite having a smaller customer base.

19.6 Validate Extreme Transactions When Operational Data Is Available

The current project retains extreme transactions because there is no independent reference source. In a production environment, unusually large transactions should be validated against:

- Order management systems
- Customer records
- Payment records
- Inventory systems
- Shipping records

This would allow the business to distinguish legitimate bulk purchases from data-quality issues.

20. Limitations

20.1 Missing Customer IDs

A significant number of transactions do not contain CustomerID. Therefore, customer-level analysis represents identifiable customers rather than every transaction.

20.2 Single Transaction Table

The project uses a single transaction dataset. There are no separate customer, product, inventory, payment, or operational tables available for independent verification.

20.3 Extreme Transactions

Extreme transactions were retained because they may represent legitimate business activity. However, they can have a significant influence on aggregated revenue.

20.4 Geographic Assignment

A small number of customers appeared across multiple countries. A consistent single-country assignment was therefore used for customer-level geographic analysis.

20.5 Historical Data

The dataset covers December 2010 to December 2011. The findings describe historical behavior and should not automatically be interpreted as current market conditions.

21. Conclusion

The Online Retail Sales Analysis transformed raw transaction-level data into a structured view of sales performance, customer behavior, product performance, and geographic contribution.

The analysis showed that revenue is strongly concentrated among high-value customers. High-Value Active customers generated approximately **64.75% of identifiable customer revenue**, while the top 10 customers contributed approximately **17.40%**.

RFM analysis provided a useful framework for distinguishing customers based on recency, frequency, and monetary value. It also identified High-Value Lapsed customers as a potential re-engagement opportunity.

Geographic analysis showed that the United Kingdom is the dominant market, while smaller markets can still contain highly valuable customers.

Product analysis identified major revenue-generating products while also demonstrating the importance of investigating unusually large transactions before drawing conclusions.

Finally, the Power BI dashboard brings these findings together into an interactive business view, allowing sales performance, customer value, product performance, and geographic contribution to be evaluated together.

Overall, the analysis demonstrates how transaction-level data can be converted into meaningful business insights through a combination of data cleaning, exploratory analysis, customer segmentation, RFM analysis, and business intelligence visualization.